

Designing Constitutional Review Institutions by Simulation

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Abstract

Constitutional review institutions face a persistent design problem: courts need enough independence to protect rights and legal continuity, but enough accountability to avoid becoming an unreviewable partisan veto point. This paper introduces a separate computational simulator for comparing supreme-court and constitutional-review designs. The model varies appointment methods, court size, term limits, removal standards, recusal rules, emergency docket procedures, voting thresholds, opinion-coalition behavior, panel versus en banc review, dual or cross-checking courts, constitutional councils, legislative override rules, and independence/accountability trade-offs. The current version adds docket subtypes, legal-domain profiles, review-path and admissibility filtering, vacancy and replacement dynamics, first-class emergency applications, override outcomes, formal and practical post-decision response models, expanded institutional-design fields, split direct outputs and derived indices, adversarial manipulation campaigns, empirical calibration source tables, seed-robustness bands, named-prior uncertainty bands, multi-family legislative-import comparison, and pairwise mechanism ablations. A data-import boundary allows legislative simulator outputs to become constitutional dockets without making this project depend on the legislative simulator source tree.

1 Motivation

The motivating question is not whether one court design is ideal in isolation. It is whether institutional designs can be compared under shared docket assumptions and explicit value tradeoffs. A court can be independent while losing democratic legitimacy, responsive while failing rights protection, stable while entrenching bad doctrine, or active while creating constitutional conflict. The simulator is built to expose those tensions rather than collapse them into a single institutional label.

The current model should be read as a design-search scaffold. It does not claim to forecast any real supreme court. It creates a synthetic filed universe, filters filings through review-path and admissibility rules, routes admitted cases through institutional designs, and reports comparative indicators. The v2 campaign and v4 diagnostics are therefore useful for identifying which mechanics deserve deeper calibration: appointment selection, emergency-docket restraints, cross-court review, council warnings, override channels, recusal discipline, replacement timing, strategic political response, legal-domain mix, intake filtering, and adversarial manipulation of procedural rules.

2 Model

Each generated world contains a pool of potential justices, filed constitutional disputes, and a legislative-output profile. Filed cases vary by legal ambiguity, rights burden, democratic mandate, partisan salience, emergency pressure, requested emergency relief, legislative quality, constitutional-conflict potential, public attention, override pressure, lower-court conflict, lower-court error risk,

and certiorari pressure. The docket separates facial challenges, as-applied challenges, election disputes, emergency stay applications, executive-power disputes, administrative-law challenges, structural cases, economic-regulation cases, and rights claims. Legal-domain profiles shift baseline case attributes so rights, election, executive-power, administrative, structural, and economic-regulation cases do not behave as interchangeable docket labels. A review-path layer distinguishes paid certiorari, IFP certiorari, discretionary certiorari, direct constitutional complaint, amparo, filtered QPC-style referral, abstract ex ante and ex post review, concrete referral, emergency application, interbranch dispute, and electoral review. Admission filtering then turns filed disputes into screened-out matters or merits transfers using SG/CSV-style signal, amicus intensity, split maturity, relists, specialist counsel, vehicle-quality defects, and conditional reversal probability; there is no preliminary fixed-size top-case selector before admission. Justices vary by ideology, independence, rights sensitivity, accountability pressure, institutionalism, partisan loyalty, and emergency deference.

Each scenario supplies a court design. The court design selects reviewers, applies recusal rules, routes cases to full-court or panel review, determines whether emergency relief occurs through an application-only path, response window, reasoned temporary stay, shadow stay, merits acceleration, or expiration without relief, applies a voting threshold, optionally invokes cross-checking or council review, and optionally allows legislative or popular override. Emergency applications carry applicant type, application class, requested relief, response request, full-court referral, status-quo effect, public disagreement risk, and merits follow-through. Override attempts can fail, succeed through ordinary supermajority action, proceed through delayed reenactment or referendum, be blocked by a rights carveout, or recur as a repeated override. Strategic actors choose whether to comply, evade, reenact, flood emergency applications, organize override campaigns, or pressure appointments. Their post-decision response is split into formal legal response and practical implementation response, including repeal, replacement, narrowed reenactment, weak-form override, amendment, court-curbing, delay, administrative substitution, symbolic compliance, bureaucratic resistance, and open noncompliance. Decisions update court state tracking precedent stability, conflict load, legislative defiance, executive emergency strategy, appointment manipulation pressure, and override adaptation. Court rosters also change over simulated time through term-limit schedules, retention pressure, life-tenure attrition, and appointment-method-specific replacement selection. Court design records now include appointment fragmentation, confirmation threshold, vacancy-deadlock risk, term renewability, retirement age, size-change difficulty, recusal consequences, quorum-failure risk, and per-remedy voting thresholds; even court sizes and senate/panel structures are permitted.

3 Legislative Input Boundary

This project remains separate from the legislative simulator. Legislative outputs are imported as CSV data and summarized into a neutral profile with enacted volume, legal quality, weak-mandate rate, rights risk, partisan skew, volatility, public legitimacy, and override pressure. This allows laws generated under legislative simulations to become inputs for constitutional review without coupling Java packages or build systems.

The v2 campaign was run with the adjacent congressional simulator’s main campaign CSV as an imported input. The importer summarized that file as enacted volume 0.343, legal quality 0.610, weak-mandate rate 0.175, rights risk 0.104, partisan skew 0.237, volatility 0.120, and public legitimacy 0.547. A separate legislative-family diagnostic compares the same import contract against seven congressional-simulator campaign families, including v0, v5, v10, v15, v20, v21-paper, and manipulation-stress report outputs.

4 Metrics

The headline metrics are legal stability, rights protection, partisan alignment, shadow-docket abuse, legitimacy, reversal rate, constitutional conflict, and democratic responsiveness. Legal stability is now also decomposed into precedent stability, statutory stability, and interbranch compliance. Emergency legitimacy risk is reported separately from shadow-docket abuse, so opacity and emergency procedure legitimacy are not collapsed into the same quantity as abusive merits-like relief. Rights-claimant success is reported both as an aggregate direct output and by legal domain for rights, structural, election, executive-power, administrative-law, and economic-regulation cases. Strategic pressure summarizes legislative defiance, executive emergency strategy, appointment manipulation pressure, override adaptation, evasion, emergency flooding, override campaigning, appointment-pressure campaigning, court-curbing, and open noncompliance. Higher legal stability, rights protection, legitimacy, and democratic responsiveness are generally better. Lower partisan alignment, shadow-docket abuse, emergency legitimacy risk, reversal rate, constitutional conflict, administrative cost, and strategic pressure are generally better. Petition, admission, screen-out, merits-transfer, invalidation, emergency grant, merits acceleration, replacement, recusal, quorum failure, concurrence, dissent, fragmentation, en banc, council warning, cross-check disagreement, override-attempt, override-success, formal-response, practical-response, and actor-choice rates are diagnostic rather than automatically good or bad.

The reported directional score is a reading aid. It averages a stability/rights score, a legitimacy/control score, rights-claimant success, elite acceptance, and administrative feasibility. Direct outputs such as rights-claimant success, doctrinal depth, remedial breadth, lower-court compliance, elite acceptance, public confidence, and coalition fragmentation are reported separately and should not be treated as automatically interchangeable components of a single welfare function.

5 V2 Campaign

The current campaign compares fourteen institutional scenarios across twenty assumption cases: the original broad cases, sensitivity sweeps for appointment capture, emergency pressure, rights risk, and democratic mandate, plus adversarial stress cases for appointment timing, emergency-application flooding, override evasion, recusal pressure, and court-expansion retaliation. The reported run used 80 runs per case, 64 filed matters per run, seed 20260501, and the imported congressional simulator campaign CSV.

Scenario	Score	Admit	Rights	Shadow	Emerg. risk	Public	Conflict
18-year staggered terms	0.588	0.560	0.613	0.068	0.190	0.539	0.364
No emergency relief without merits review	0.585	0.582	0.617	0.004	0.187	0.602	0.367
Stylized current U.S.-like court	0.584	0.560	0.618	0.235	0.255	0.410	0.376
Nonpartisan commission	0.584	0.578	0.616	0.070	0.197	0.557	0.370
60 percent invalidation threshold	0.583	0.561	0.602	0.061	0.200	0.525	0.364
Panel with en banc correction	0.581	0.582	0.614	0.071	0.196	0.604	0.371
Comparative 16-seat constitutional senates	0.570	0.562	0.597	0.062	0.200	0.593	0.363
Cross-checking constitutional court	0.551	0.577	0.595	0.063	0.205	0.551	0.400

Table 1: Selected v2 campaign averages after replacing preliminary top-case selection with filed-universe admission filtering. Lower values are preferred for shadow-docket abuse, emergency legitimacy risk, and conflict. The full CSV and Markdown report are generated under `reports/`.

6 Early Interpretation

The v2 results shift after making admissibility operate on the full filed universe. Term regularization now leads the directional score by combining low shadow-docket abuse, moderate emergency legitimacy risk, and relatively low conflict. The no-emergency-relief-without-merits design remains the strongest emergency-docket reform on shadow abuse, emergency legitimacy risk, and public confidence, even though it pays administrative cost through merits acceleration. The stylized current U.S.-like court has high rights protection in this run, but much higher shadow-docket abuse and emergency legitimacy risk. Cross-checking and dual-court designs reduce partisan alignment, but the current disagreement-resolution model still increases constitutional conflict and administrative cost. The comparative 16-seat constitutional-senate scenario confirms that even court sizes and chamber-like structures can be modeled, but it does not automatically dominate simpler reforms.

The calibration run now passes all seventeen current guardrails for docket mix, invalidation, recusal frequency, emergency-docket abuse, split legal stability, partisan alignment, override attempts, and admissibility-aware merits transfer. The guardrails are computed from 758 normalized source observations derived from SCDB 2025 Release 01, the Supreme Court Shadow Docket Database, Black and Epstein’s recusal data, and the Deep Research synthesis CSV tables, with source-specific tolerances for simulator concepts that do not map one-to-one to historical tables [6, 4, 5, 1, 7]. The generated appendix reports raw source minima, percentiles, medians, maxima, terms, and source keys. Seed robustness is tight under the current deterministic averaging setup, suggesting that headline ordering is not being driven by a single random stream. Named-prior sweeps introduce wider uncertainty bands by varying polarization, appointment capture, emergency pressure, rights risk, weak mandate, public pressure, conflict, and imported legislative profiles. The legislative-family comparison imports seven congressional-simulator report families. The mechanism ablation report shows that emergency restraint, term regularization, and supermajority invalidation thresholds reduce shadow-docket abuse relative to the stylized current-court baseline, while cross-checking and dual-court filters reduce partisan alignment at the cost of conflict and administration. The manipulation-stress campaign preserves the same broad tradeoff structure but lowers scores across the board, especially under emergency flooding and override-evasion assumptions.

7 Remaining Work

The next increments should focus on replacing normalized summaries and simple policies with stronger empirical and strategic validation. The highest-value additions are:

1. automate raw-data refresh for SCDB, Harvard Law Review statistics, shadow-docket data, and recusal data rather than relying on checked-in normalized tables;
2. convert named priors into sampled prior distributions with Monte Carlo or Bayesian sensitivity summaries;
3. let legislature and executive actors learn from repeated outcomes rather than using a static strategic policy;
4. calibrate review-path-specific admission denominators against raw certiorari, amparo, constitutional complaint, QPC, abstract-review, and concrete-referral data;
5. validate the legislative import contract against raw legislative simulator run outputs, not only published aggregate campaign CSVs.

8 Reproducibility

The simulator is dependency-free Java and is run through `make`. The core commands are:

```
make test           make campaign           make diagnostics
make parameter-sweep make legislative-family-comparison make paper
```

The v2 campaign and v4 diagnostic artifacts are written under `reports/`. The preserved v0 and v1 campaigns can still be regenerated with `make campaign-v0` and `make campaign-v1`. The paper is built from this file with `make paper`. Normalized calibration source observations live under `data/calibration/`. The software and campaign outputs are treated as the primary artifacts [3]. Imported legislative profiles are optional and come from the separate legislative simulator’s report CSV contract [2].

References

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